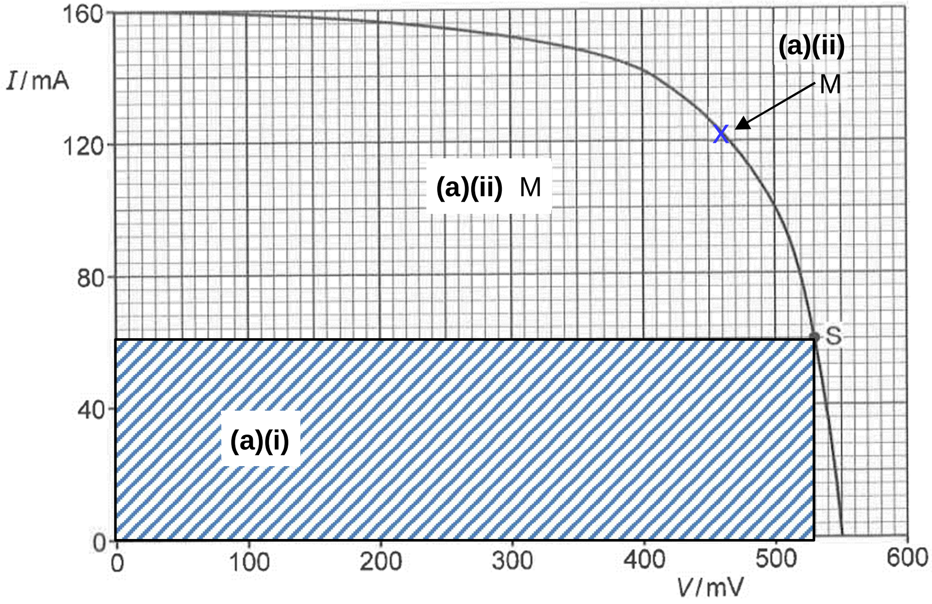
	≈ 28
6(a)(i), (ii)	 **Fig. 6.2**

No.	Solution
6(b)(i)	From Fig. 6.2, $V = 500$ mV when $I = 500$ mA , $R = \dfrac{V}{I}$ $= \dfrac{500 \times 10^{-3}}{100 \times 10^{-3}}$ $= 5.0\ \Omega$
6(b)(ii)	Power dissipated in resister R $P = I^2 R$ $= \left(100 \times 10^{-3}\right)^2 \times 5.0$ $= 0.050$ W
6(b)(iii)	From Fig. 6.2, $V = 550$ mV when $I = 0$ mA , i.e. e.m.f. of solar cell, $E = 550$ mV When $I = 500$ mA , $E = V + Ir$ $r = \dfrac{E - V}{I}$ $= \dfrac{(550 - 500) \times 10^{-3}}{100 \times 10^{-3}}$ $= 0.50\ \Omega$
7(a)	similarity: